

System Architecture Diagrams

Visual companion to `docs/ARCHITECTURE.md` (which follows the HLD v1.0). All diagrams are Mermaid — they render on GitHub and in any Mermaid-capable editor. Diagrams reflect the **implemented** code (paths shown per diagram).

[FastAPI orchestration](#)[React PWA](#)[DPI rails • ULI / AA / Land / DISCOM](#)[BRI 0–100 scoring](#)[Pass-through by design](#)[Consent before data \(DPDP\)](#)

Source of truth chain: HLD Document → ARCHITECTURE.md → ARCHITECTURE_DIAGRAMS.md → code

SetuCredit Core Team

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CONFIDENTIAL

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Page numbers refer to this PDF. Diagrams render from the Mermaid source in `docs/ARCHITECTURE_DIAGRAMS.md` — regenerate the PDF after editing that file.

1 System context (C4 level 1)

SetuCredit is a middleware orchestration layer: it holds no borrower data and makes no lending decision. It assembles a consented, pre-underwritten package from DPI rails, scores it (BRI 0-100), and hands it to a partner NBFC.

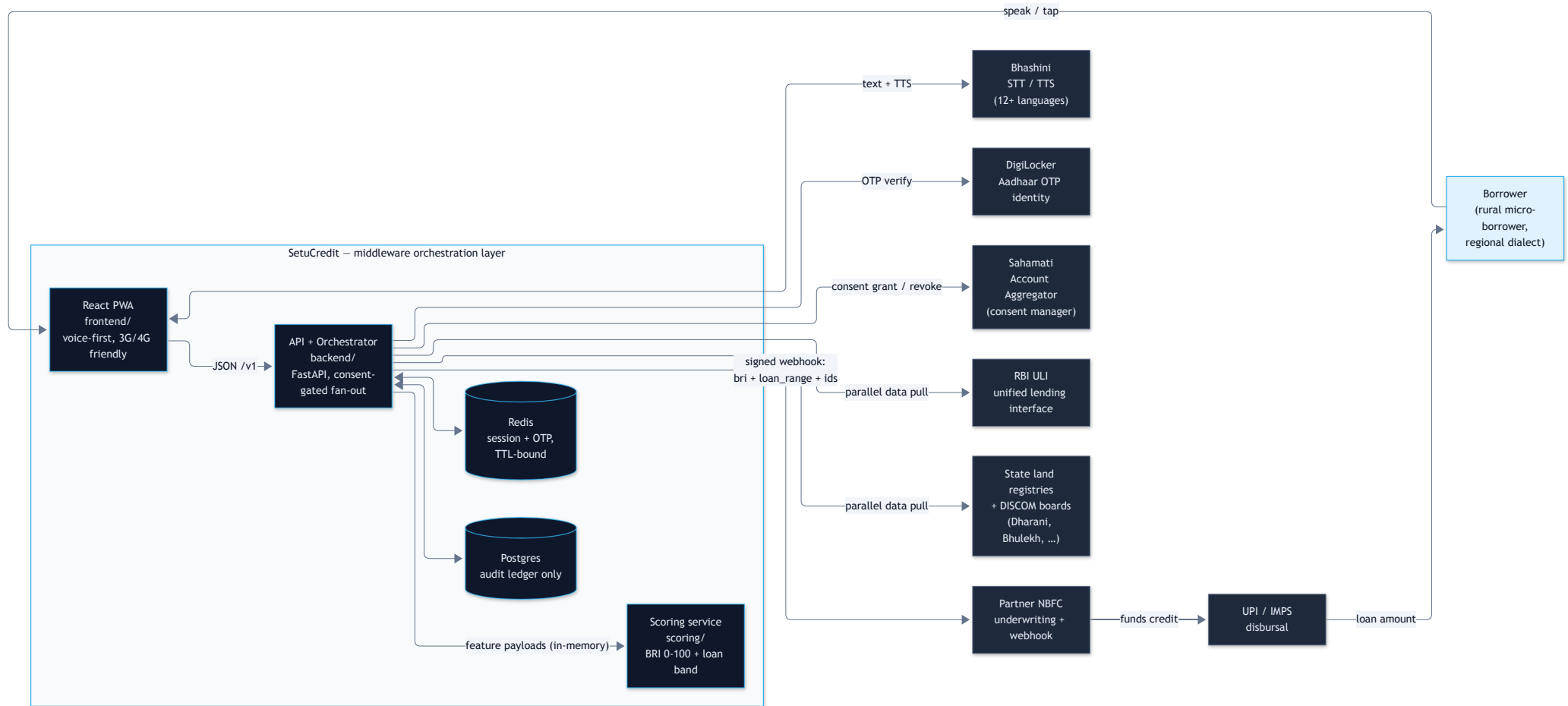
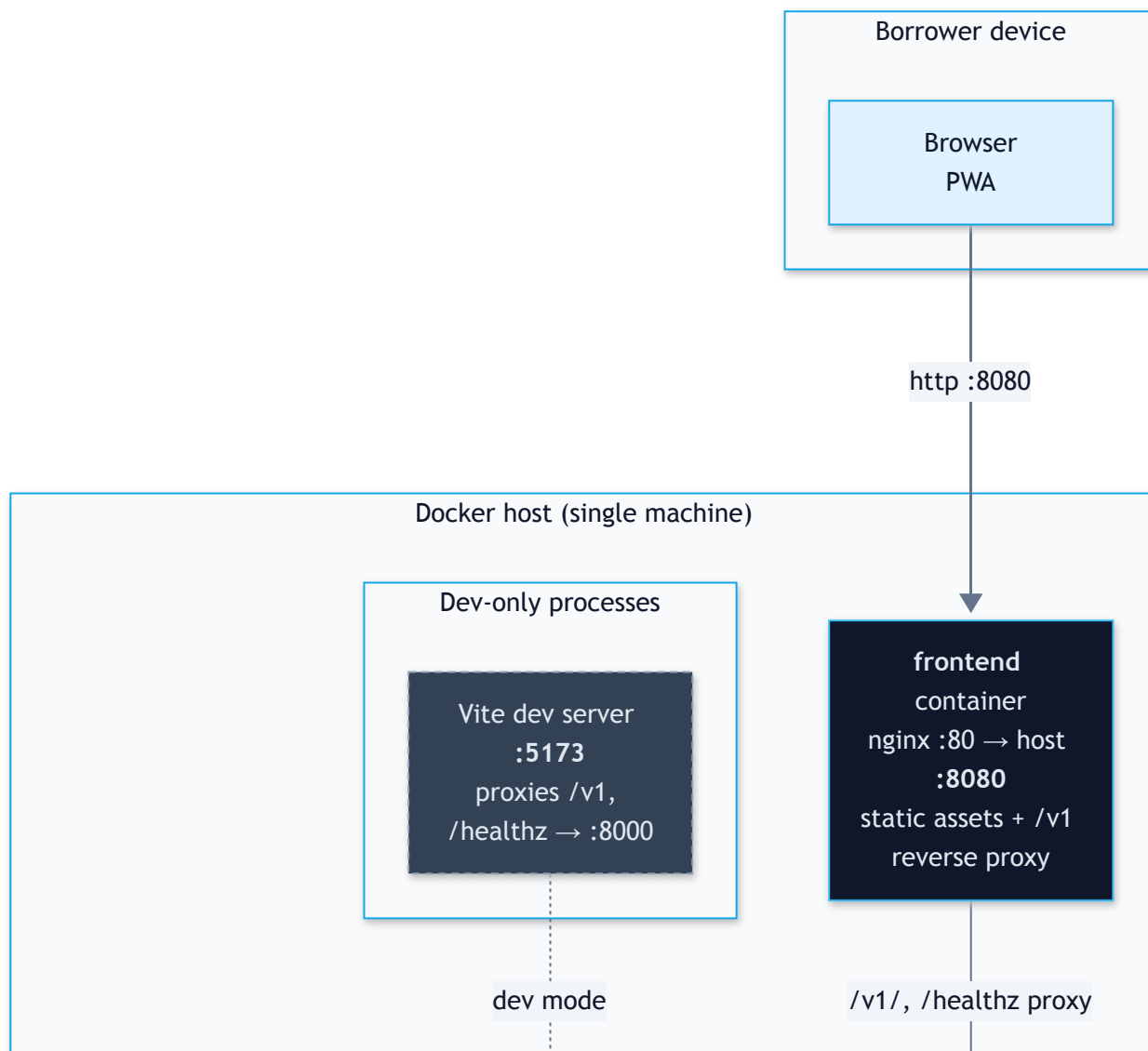
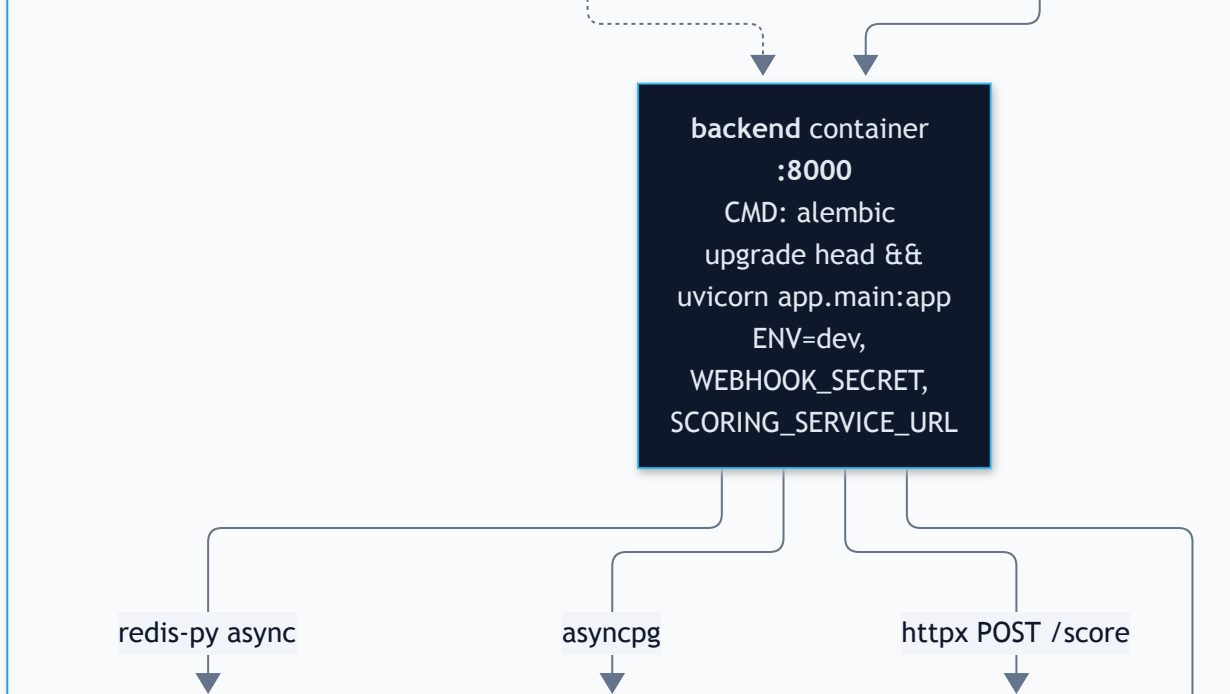


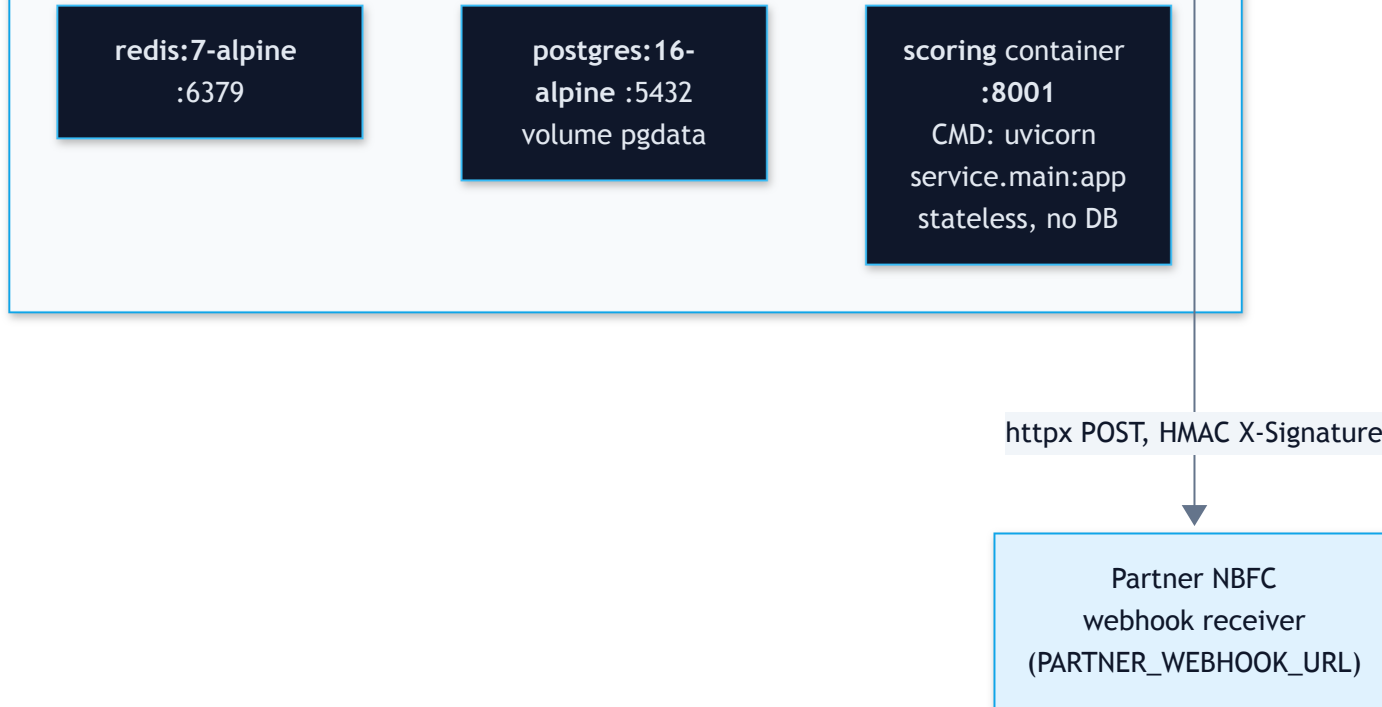
FIGURE 1 – SYSTEM CONTEXT (C4 LEVEL 1)

Two invariants (HLD §5) govern every arrow above: borrower non-financial data is never persisted (§8), and no DPI pull happens without a valid, unexpired, unrevoked consent (enforced in `backend/app/consent/service.py` → HTTP 403).

Single-host Docker Compose (`infra/docker-compose.yml`), profile `app` :







— FIGURE 2 — DEPLOYMENT TOPOLOGY

- Only **frontend :8080** and **backend :8000 / scoring :8001** are reached from outside in a demo; Postgres/Redis/scoring are on the compose network.
- Scale-out path (post-hackathon): orchestrator and scoring are already separate processes; backend can be replicated behind a load balancer (sessions live in Redis, not in process).
- Non-functional (HLD): PWA stays light for 3G/4G; the fan-out is latency-sensitive hence parallel; voice degrades to text in the same language.

3 Backend component map

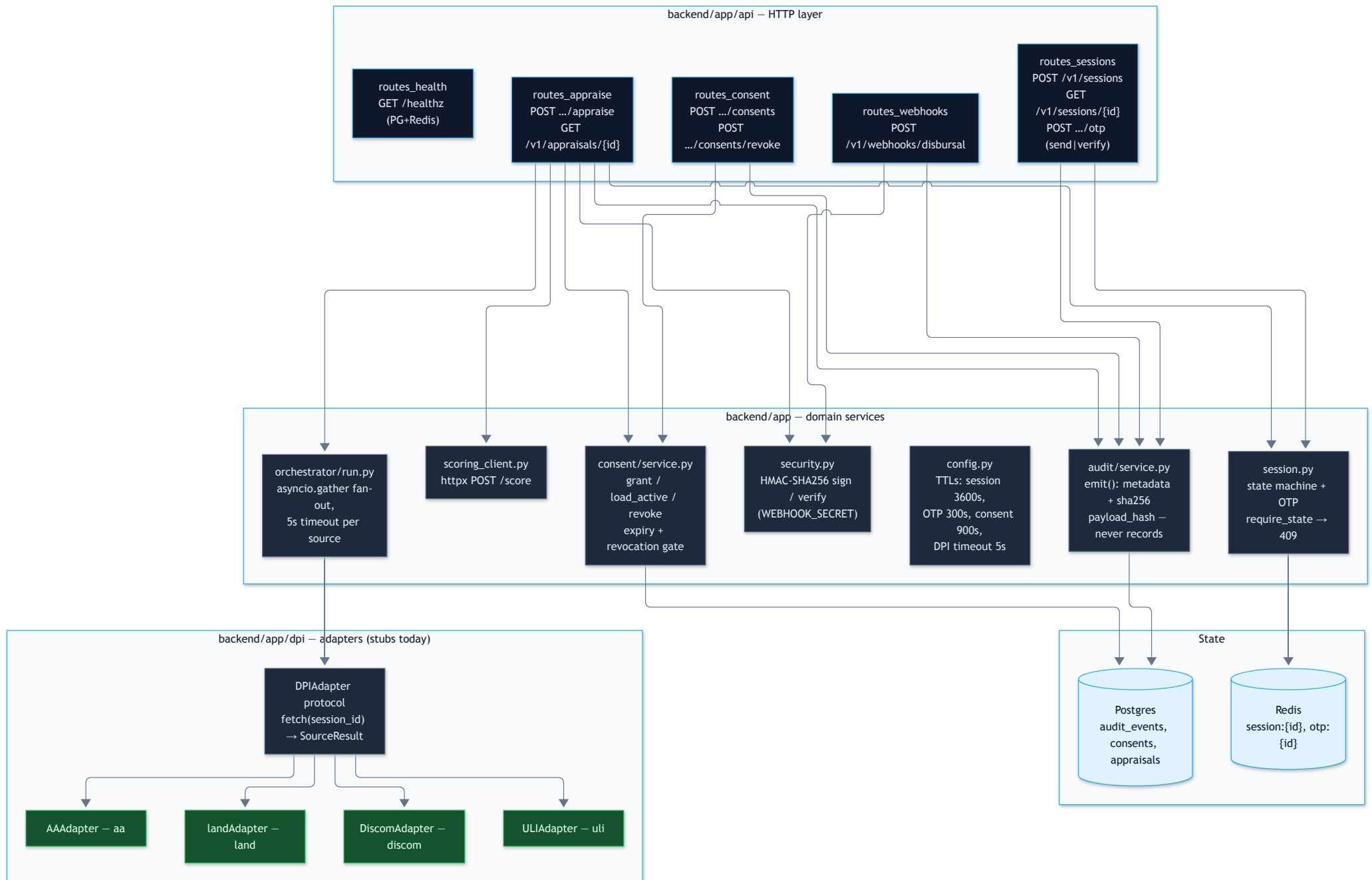
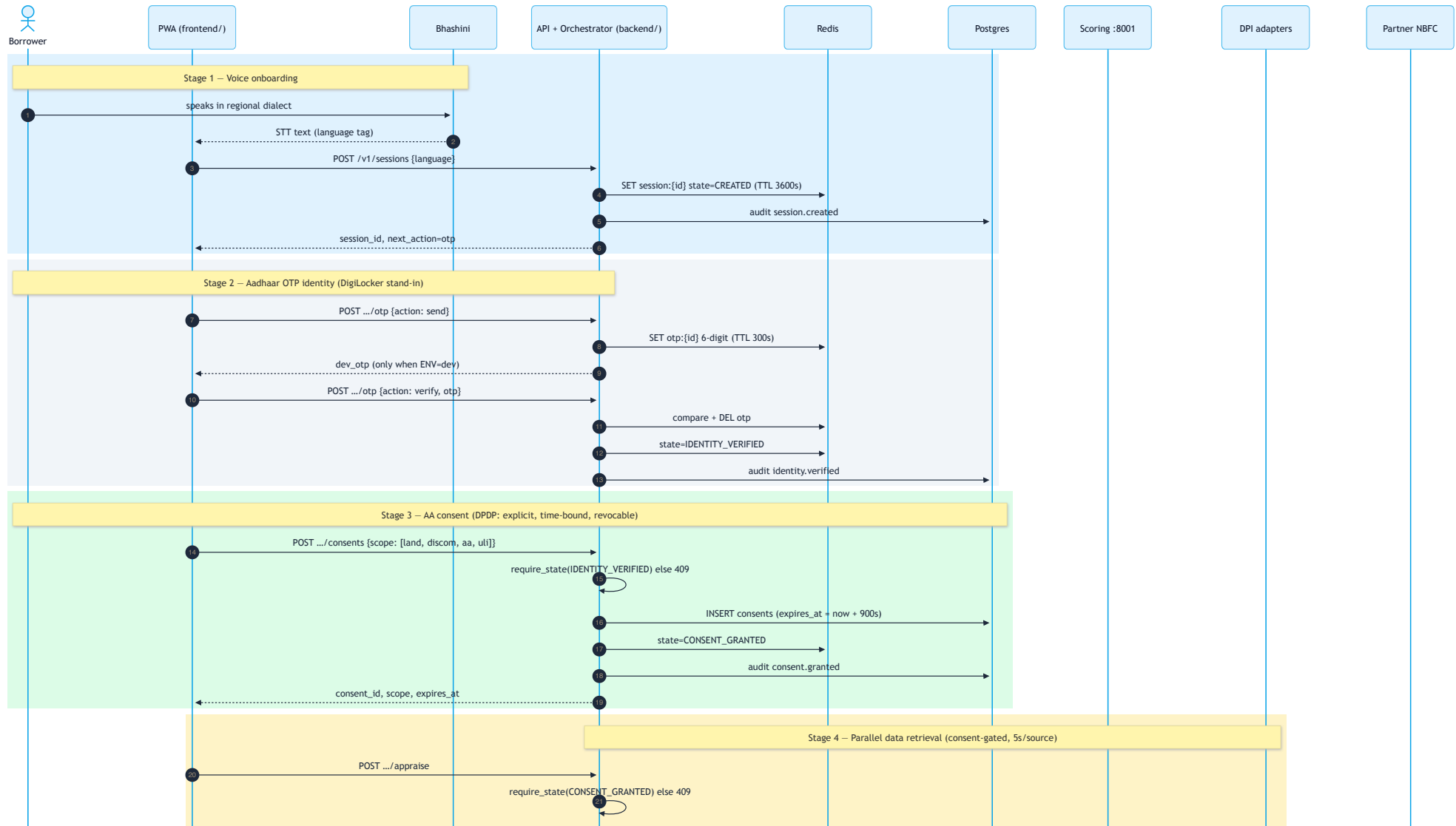


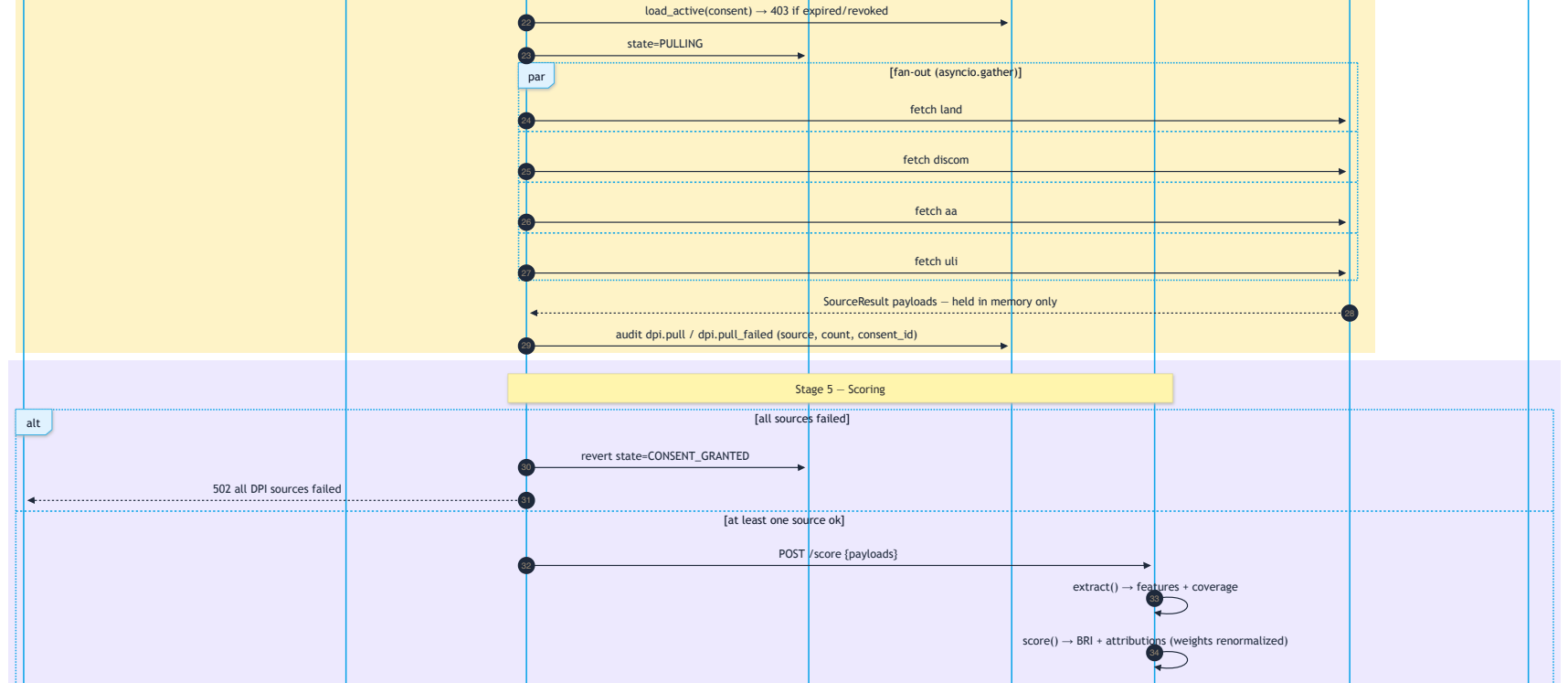
FIGURE 3 – BACKEND COMPONENT MAP

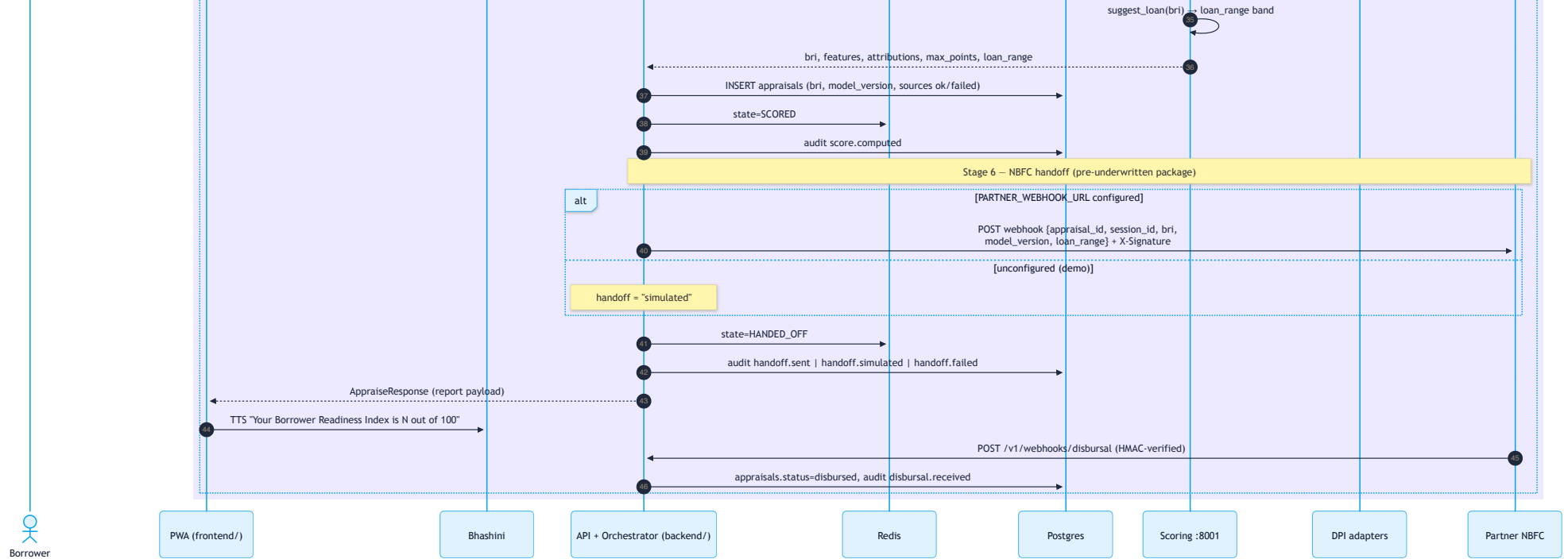
Key control points:

- **require_state** (session.py:67) — every route declares its legal source states; violations raise `StateError` → HTTP 409 (`main.py` handler).
- **Consent gate** (routes_appraise.py:31-34) — `load_active()` re-checks scope, expiry and revocation from Postgres right before the pull; failure → 403 and the session reverts to `CONSENT_GRANTED` .
- **Adapter isolation** (orchestrator/run.py:15-28) — a timeout or exception in one adapter becomes `SourceResult(ok=False)` ; one bad rail never fails the appraisal.
- **Adapters are the only entry point for raw borrower records** — nothing downstream of `orchestrator` sees raw payloads except `scoring` (as features, in memory).

HLD §3 stage numbering: 1 voice → 2 identity → 3 consent → 4 parallel pull → 5 scoring → 6 handoff/disbursal.







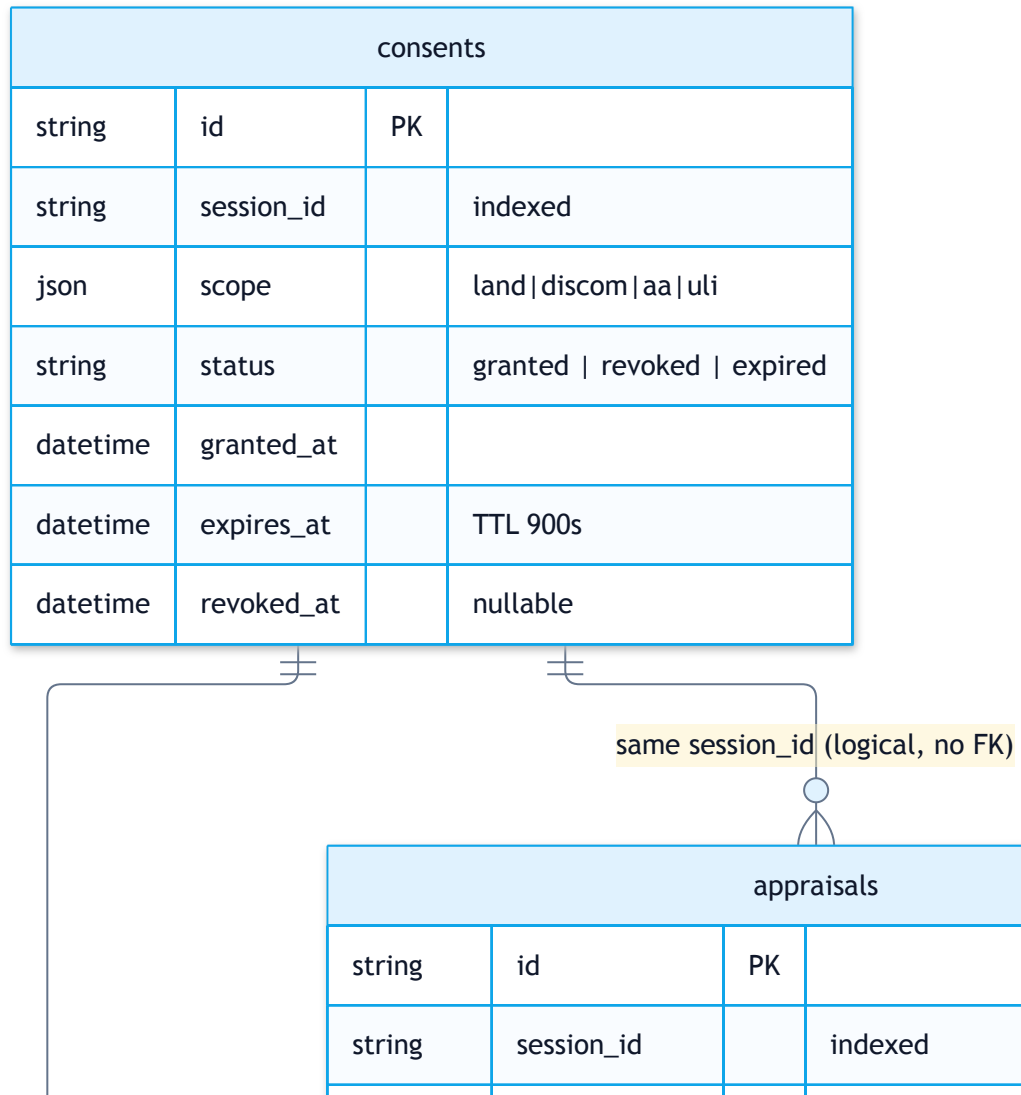
— FIGURE 4 — END-TO-END APPRAISAL SEQUENCE (6 STAGES)

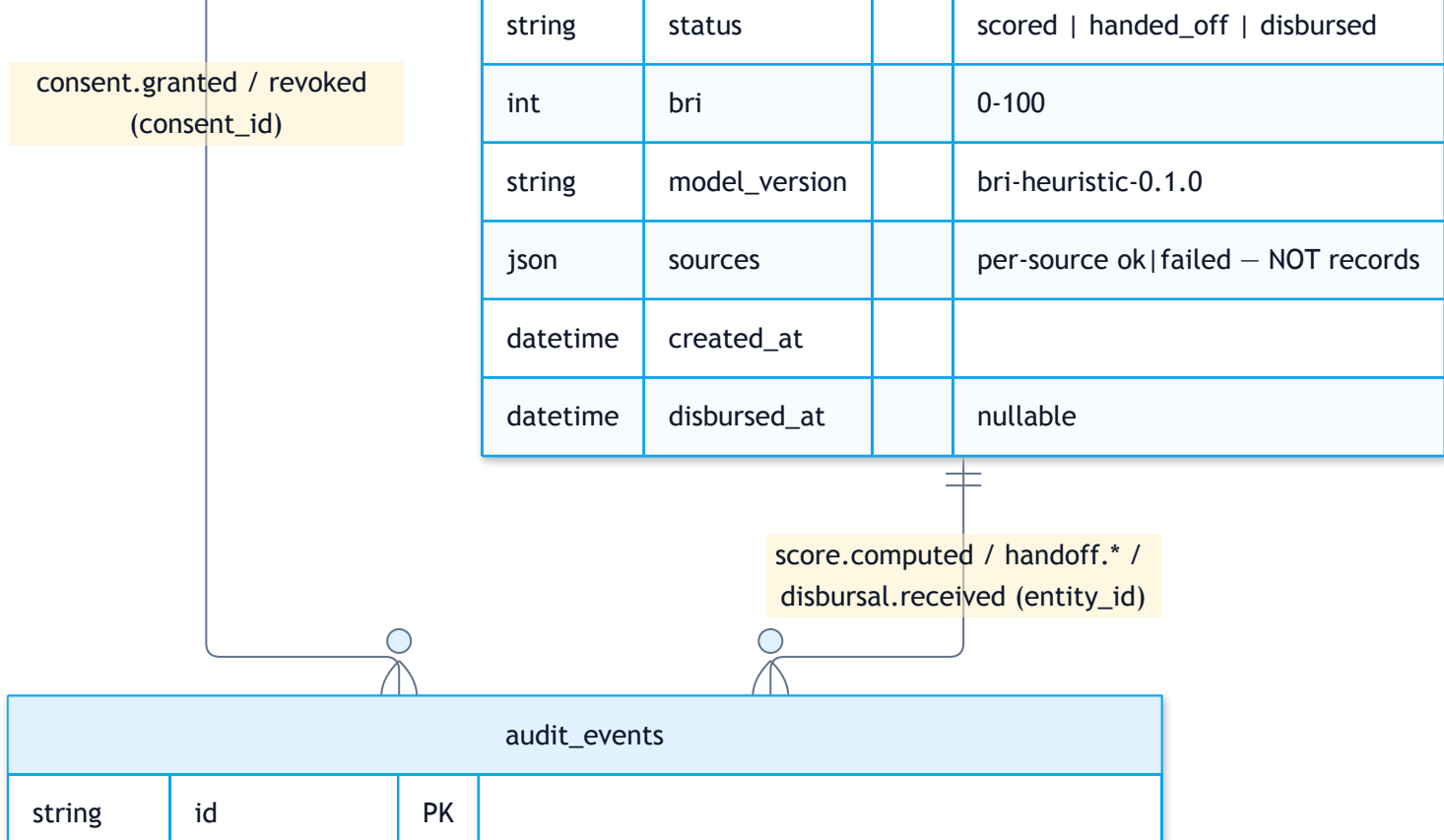
Notes:

- Scoring unavailable → 502 , session also reverts to `CONSENT_GRANTED` (routes_appraise.py:54-59).
- A failed *outbound* handoff leaves the session in `SCORED` (only `!= "failed"` advances to `HANDED_OFF`), so it can be retried; the response still returns `handoff="failed"` .
- No audio is ever stored: STT/TTS happen in the PWA; the API sees text only.

6 Data model and retention

Postgres — audit ledger only (backend/migrations/versions/508ab15e2977_...)





string	session_id		indexed
string	event_type		session.created, identity.verified, consent.granted revoked, dpi.pull dpi.pull_failed, score.computed, handoff.*, disbursal.received
string	entity_id		nullable – appraisal id
string	consent_id		nullable
string	source_id		nullable – land discom aa uli
int	record_count		nullable – count only
string	model_version		nullable
string	payload_hash		sha256 of webhook body
datetime	created_at		

— FIGURE 6 – DATA MODEL AND RETENTION

Redis – ephemeral session state (backend/app/session.py)

Key	Value	TTL
<code>session:{session_id}</code>	JSON: state, language, consent_id, appraisal_id	3600 s
<code>otp:{session_id}</code>	6-digit code (single use)	300 s

Never stored anywhere (pass-through by design): land records, utility bills, bank transactions, AA/ULI payloads, derived feature values, audio. `loan_range` is computed in-request and returned to the PWA / sent to the NBFC, but is not written to the ledger.

7 Scoring pipeline

Separate, stateless process (`scoring/`): in-memory payloads over HTTP in, BRI out.

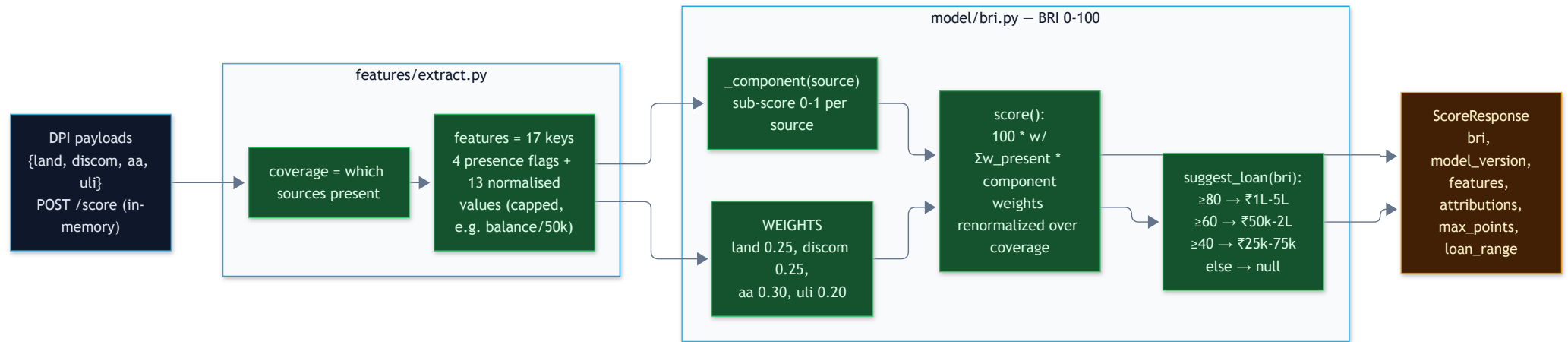


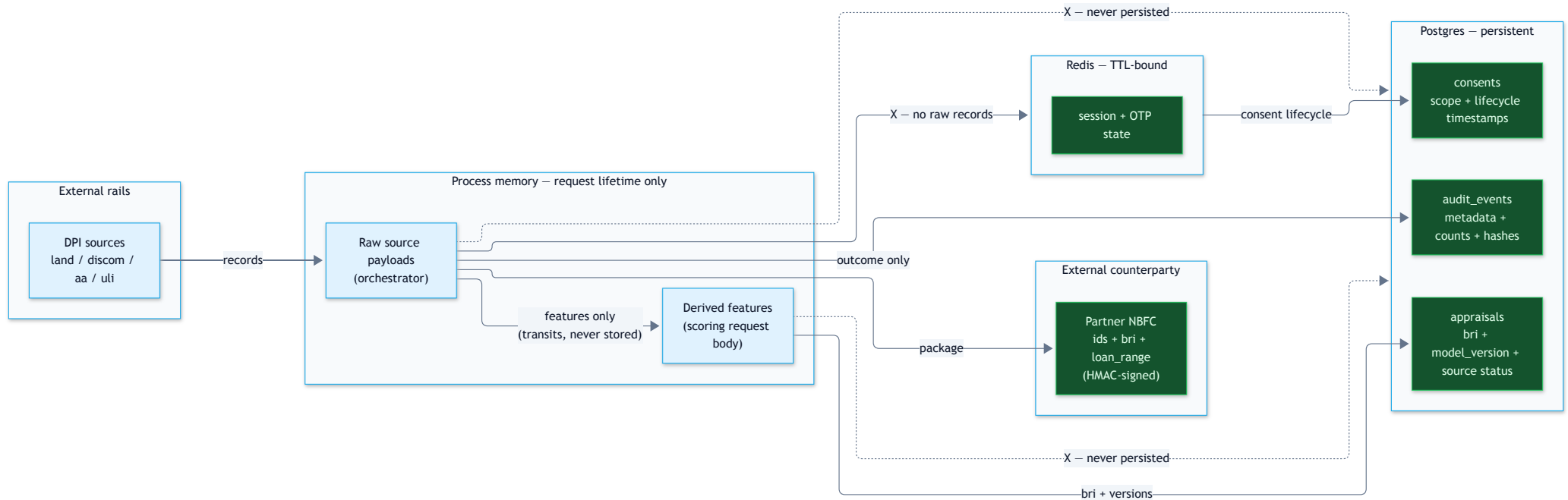
FIGURE 7 — SCORING PIPELINE

Per-source sub-scores (`_component` , `bri.py:35-62`), all clamped to [0, 1]:

Source	Formula
land	$0.50 \cdot (\text{acres}/5) + 0.30 \cdot \text{clear_title} + 0.20 \cdot (\text{years}/20)$
discom	$0.60 \cdot \text{on_time_ratio} + 0.20 \cdot (\text{months_paid}/24) + 0.20 \cdot \text{bill_level}$
aa	$0.40 \cdot \text{balance_level} + 0.40 \cdot \text{inflow_consistency} + 0.20 \cdot (\text{months}/60) - 0.30 \cdot \text{emi_ratio}$
uli	$0.50 \cdot (1 - \text{loans}/4) + 0.30 \cdot (\text{repay_months}/24) + 0.20 \cdot (1 - \text{delinquencies}/3)$

- Sparse consents still score: absent sources drop out and the remaining weights renormalize ($\sum \text{max_points} = 100$).
- `score()` and `suggest_loan()` are pure functions — the HLD's trained XGBoost model drops in behind the same interface; `model_version` records which ran.
- BRI bands for the report: **≥80 Strong**, **≥60 Good**, **≥40 Fair**, **<40 Building** (loan range absent below 40).

What may cross each boundary, and what may not — the invariant from HLD §5 rendered as a data-flow diagram:

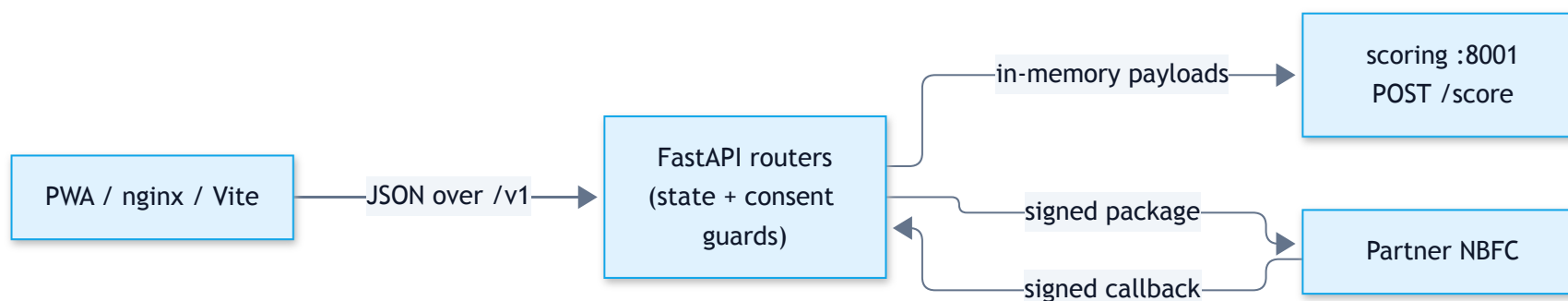


— FIGURE 8 — PASS-THROUGH TRUST BOUNDARY

Compliance posture: purpose limitation (credit appraisal only), retention = audit ledger + consent records only, revocable consent aborts any in-flight pull (DPDP); TLS 1.3 in transit, AES-256 for stored audit logs (HLD §5); webhooks to/from the NBFC are HMAC-SHA256-signed (`security.py`) with the appraisal body hashed into the ledger (`payload_hash`).

9 API surface

Method & path	Handler	Prerequisite state	On violation
POST /v1/sessions	routes_sessions	—	—
GET /v1/sessions/{id}	routes_sessions	session exists	404
POST /v1/sessions/{id}/otp	routes_sessions	CREATED	409 / 400 (bad OTP)
POST /v1/sessions/{id}/consents	routes_consent	IDENTITY_VERIFIED	409 / 422 (bad scope)
POST /v1/sessions/{id}/consents/revoke	routes_consent	any post-consent state	409 / 404
POST /v1/sessions/{id}/appraise	routes_appraise	CONSENT_GRANTED + live consent	409 / 403 / 502
GET /v1/appraisals/{id}	routes_appraise	appraisal exists	404
POST /v1/webhooks/disbursal	routes_webhooks	valid X-Signature	401 / 400 / 404 / 422
GET /healthz	routes_health	PG + Redis ping	503 if degraded
POST /score (internal :8001)	scoring/service	—	—
GET /healthz (internal :8001)	scoring/service	—	—



— FIGURE 9 — API SURFACE

Source of truth: `docs/SetuCredit_HLD_Document.pdf` → `docs/ARCHITECTURE.md` → this file → code. Keep all four in sync; where they conflict, the HLD wins.